	Design Development Plan Form		
Project Name	Some Project		
Project Number	Ryymmxxx		
Document Author	D. Engineer		
File Name	Ryymmxxx_CAL001 Some Project	**Issued Date**	27/04/2026

1 Document Purpose

This is a Quarto document template (`.qmd`) intended to be used to develop engineering calculations in the format of a typical engineering calculation pad.

To use, install quarto and then run the following command to download the template:

```
quarto use template skane88/quarto-engformat/engformat_03
```

You can learn more about creating custom Typst templates here:

https://quarto.org/docs/output-formats/typst-custom.html

For actual use, the introduction should be a *brief* description of the calculation objective and scope and ideally include a picture.

QA Record

Originator	D. Engineer	**Checked**	D. Engineer	**Approved**	D. Engineer	27/04/2026

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2 Basis of Design

2.1 Design Brief & Client Requirements

2.2 Design Constraints and Battery Limits

2.3 Parameters

2.3.1 Geometry

2.3.2 Material Properties

2.4 Design Changes

2.5 Governing Standard and Regulations

The following table lists the primary standards and specifications and specifications followed for this project.

The following is a pipe-table. This is the simplest form of table with limited formatting options. It can be cross referenced (Table 1), and can be given captions.

Table 1: Standards and Specifications

Item	Reference	Year	Description
	AS 4100	2020	Steel Structures

The following is a list-table (Table 2). This is a complex table type that allows more formatting options and the ability to merge cells. However, the table is quite difficult to write, so use should generally be limited.

Table 2: Standards and Specifications.

Item	Reference	Year	Description
Standards			
	AS 4100	2020	Steel Structures
Specifications			

It is also possible to directly incorporate typst blocks into the document for complete control over formatting.

3 Work Records

The following design calculations have been relied upon in the preparation of this calculation:

Table 3: Work Records

Doc Number	Title	Revision	Link / Path
R26xxx_CAL002	Structural Steel Design	A	link
R26xxx_CAL003	Foundation Design	A	link

4 Design Loading

This section describes the loads applied.

4.1 Gravity Loads

Document gravity loads here.

4.1.1 GSW Structural Self Weight

Structural self weight is applied as an acceleration in the model unless noted otherwise.

4.1.2 GFM Flooring

Floor mesh self weight is assumed to be:

$$mass = 50\,\mathrm{kg/m^2}$$

$$GFM = mass \times g_{acc} = 0.5\,\mathrm{kPa}$$

Flooring is applied as follows:

- A dummy load case of 1kPa is applied to all floor areas.
- The actual floor load is then applied as a load combination of 0.5 x the dummy case.

4.1.3 GHR Handrails

Handrails are assumed to be 15kg/m.

$$GHR = 15\ \mathrm{kg/m} \times g_{acc} = 1.5\ \mathrm{kN/m}$$

4.1.4 GME Mechanical Equipment

Break into multiple load cases if required.

4.2 Live Loads

4.2.1 QLL5 Standard Floor Live Load

4.2.2 QME Mechanical Equipment Live Load

Break into multiple load cases if required.

4.3 Load Combinations

5 Analysis Model

The following figure shows the analysis model:

5.1 Restraints, Releases & Offsets

The following restraints, releases and offsets have been used:

- All columns assumed to be pinned at base unless otherwise noted.
- All members assumed to be pinned at their ends unless otherwise noted.
- Members modelled at their centrelines unless otherwise noted.

5.2 Analysis Method

The following analysis methods have been used:

- Non-linear static analysis.
- Linear buckling where required to determine effective lengths for members.

5.3 Steel / Concrete Design Assumptions

The following steel and concrete design assumptions have been used:

- …

6 Summary

We have done some awesome calcs.

7 Calculation

Copy this section as required.

7.1 Example Calcs

Do some calculation here.

```
from cycler import cycler
import numpy as np
import polars as pl
import matplotlib.pyplot as plt

a = 1
b = 2
a + b
```

```
3
```

And now plot a graph (see Figure 1):

```
no_points = 101
jitter_scale = 0.2
x = np.linspace(0, 2 * np.pi, no_points)
y1 = np.sin(x) + (np.random.random(no_points) - 0.5) * jitter_scale
y2 = np.cos(x) + (np.random.random(no_points) - 0.5) * jitter_scale
y3 = np.sin(x * 2) + (np.random.random(no_points) - 0.5) * jitter_scale
y4 = np.cos(x * 2) + (np.random.random(no_points) - 0.5) * jitter_scale
y5 = np.sin(x * 3) + (np.random.random(no_points) - 0.5) * jitter_scale
y6 = np.cos(x * 3) + (np.random.random(no_points) - 0.5) * jitter_scale
y7 = np.sin(x * 4) + (np.random.random(no_points) - 0.5) * jitter_scale
y8 = np.cos(x * 4) + (np.random.random(no_points) - 0.5) * jitter_scale
```

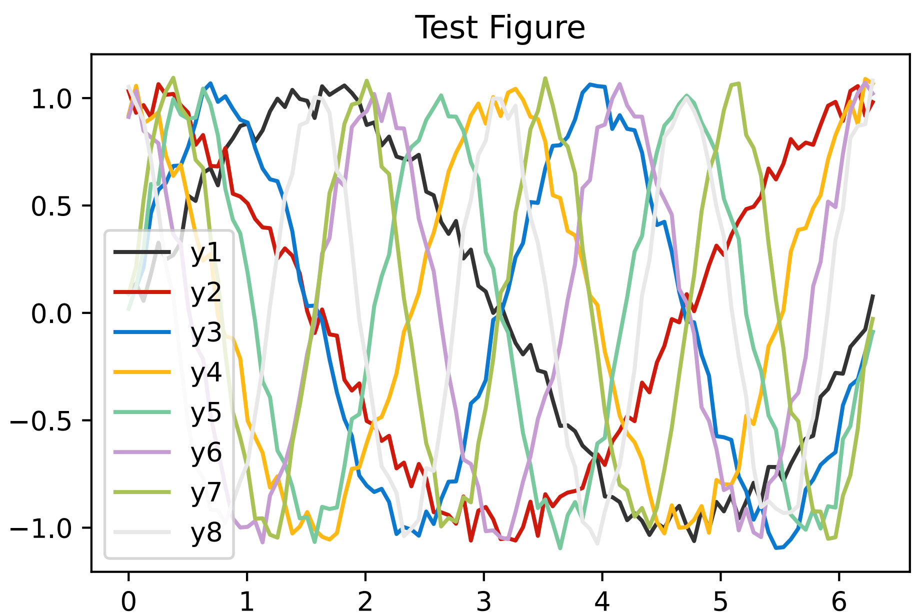

Figure 1: Test Figure

Woohoo!

And do some data processing with Polars (see Table 4):

```
df = pl.DataFrame({"x": x, "y": y1, "z": y2})

pl.Config.set_tbl_hide_column_data_types(True)
pl.Config.set_float_precision(3)

df.head()
```

Table 4: Test Data

x	y	z
0.000	0.093	1.033
0.063	0.137	0.931
0.126	0.056	0.967
0.188	0.181	0.910
0.251	0.327	1.065

Tip 1: Tip

This is an example of a callout “tip”.

Warning

This is an example of a callout “warning”.

! Important

This is an example of an “important” callout.

Callouts can even be cross-referenced - see Tip 1. So can sections - see **?@sec-intro**.

8 Appendix A

9 Appendix B